NATIONAL INSTITUTE OF TECHNOLOGY PATNA
MID-SEMESTER EXAMINATION, MARCH 2025

Program: B.Tech. (CSE)
Department: CSE
Full Marks: 30

Semester: 2nd
Course Code: CS24107
Course Name: Web Technology
Duration of Examination: 2 hours

INSTRUCTIONS

- Answer all the questions. Assume Missing data, if any.
- Students are requested to write the Section on the top of the answer book.

Q1. **[1x6 = 6M][CO3][L2]** Will the following JS code snippet run? If Yes, provide the output, if NO, explain the reason along with correction required (No marks if proper reasoning not provided):

a. `console.log(a);` `var a = 5;`	b. `console.log(a);` `let a = 10;`
c. `{var a = 10;}` `console.log(a);`	d. `{let x = 20;}` `console.log(x);`
e. `const a = 5; a = 10;` `console.log(a*2);`	f. `const a = Your Institute Roll Number;` `a >>> 3; console.log(a);`

Q2. **[4M][CO2] [L3]** Write a JavaScript code using events to increase the size of the image when someone moves the mouse over the image and the image goes back to its normal size when mouse is moved away from the image.

OR

Explain with illustrative example, How come multiple users of NIT Patna sees their public IP same. Explain how external servers/users are able to send data to different users of NIT Patna when they see all requests coming from the same public IP.

Q3. **[8M][CO3][L3]** Solve the following question:

a. Set the background-color to red, of any <input> element that are in focus.
b. Set a border for <img> elements that have a title attribute *ending* with the word flower (not flowers).
c. Set the background color to "red" for <a> elements that have a target attribute.
d. Set the background-color to red, when you mouse over elements with the class "master".
e. Collect user-name through a pop up box and display the collected user name in a pop-up box.
f. Style all elements with a class attribute value that contains "te".
g. Write a CSS code that makes the selected text red on a yellow background.
h. Write a CSS code that shows a <p> element when a element is Hover.

Q4. **[6M][CO1][L3]**You have sub-netted your class C network 200.138.1.0 with a subnet mask of 255.255.255.252. Please list the following: number of networks, number of hosts per network, the full range of the first three networks, and the usable address range from those first three networks. Additionally, identify the broadcast addresses for each network.

Q5. **[3M][CO3][L3]**Write a JS function to get the youngest person from following JS construct. Program should be generic in nature, so that it works for any similar construct:

```
const people = [
 { name: "Ram", age: 38 },
 { name: "Shyam", age: 35 },
 { name: "Mohan", age: 21 }
];
```

Q6. **[3M][CO3] [L3]** Write the HTML code to create the following table.

Name		**Roll Number**	**Grade**
First Name	**Last Name**		
AAA	ZZZ	1001	B
BBB	YYY	1002	A